

2. OOGENESIS

This is the formation of gametes in the female sex. It takes place in the ovaries and allows the formation of female gametes, oocytes, from germline stem cells or oogonia.

1.REMINDERS ABOUT THE FEMALE GENITAL SYSTEM

The female genital system includes internal genital organs (two ovaries, two fallopian tubes, the uterus and the vagina) and external genital organs (the vulva = the labia majora, the labia minora and the clitoris).

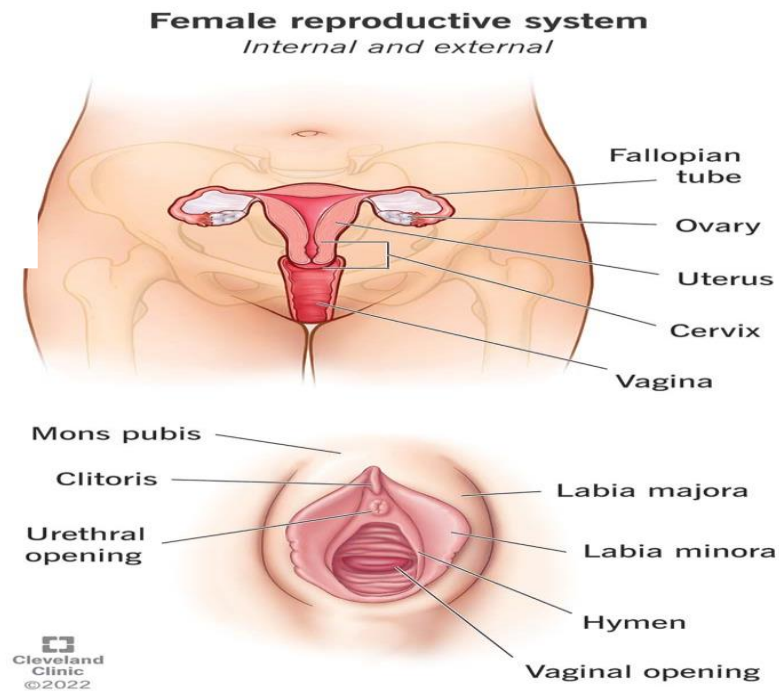


Figure 1. Female reproductive system

The vagina is a tube-shaped organ, 10 to 15 cm long, very extensible, in which spermatozoa are deposited during sexual intercourse; it is also the natural passageway of the fetus during childbirth. It is lined by a non-keratinized squamous epithelium.

The uterus is a smooth muscular organ of about 7 cm, triangular in shape, in which the embryo and then the fetus develop. It is hollowed out by a thin cavity: the uterine cavity.

The uterine lining or endometrium is the inner most part of the uterine wall, which includes a muscular layer, the myometrium, and a serous layer, the inner layer of the peritoneum.

The endometrium is composed of a lining epithelium and an underlying chorion, separated by a basal lamina:

- The epithelium is of simple prismatic type, with ciliated cells and non-ciliated cells,
- The chorion is a loose connective tissue, relatively poor in fibers and highly vascularized by spiral arterioles; it contains coiled tubular glands, which appear as invaginations of the lining epithelium.

In the thickness of this mucosa, two zones can be distinguished: the functional zone corresponding to the superficial zone, containing the glands, the thickest, and the residual zone, underlying, in contact with the myometrium.

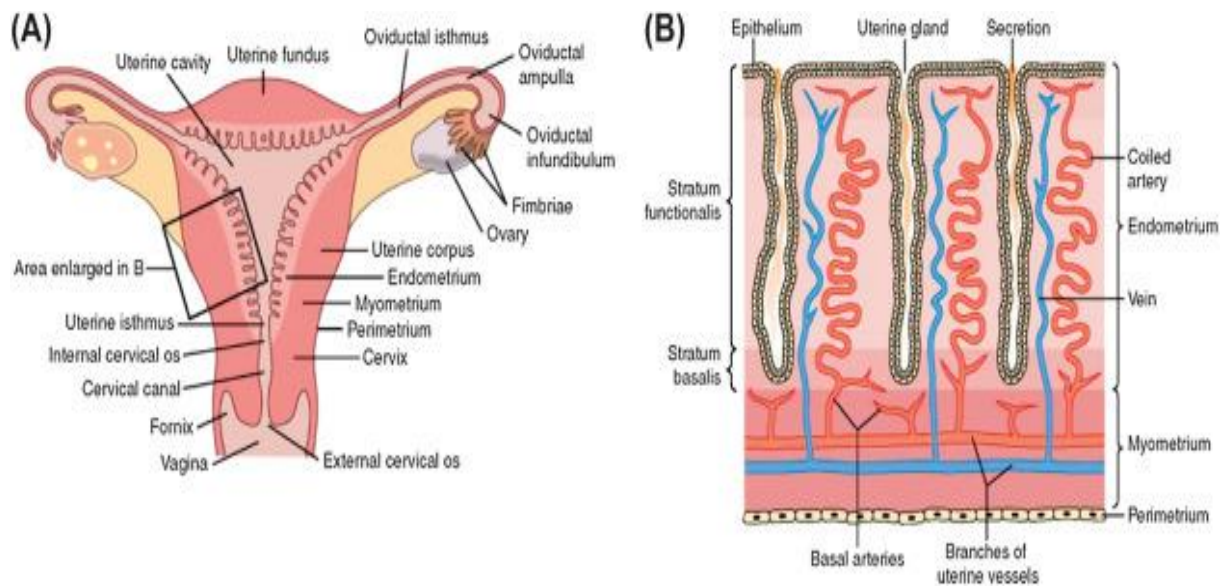


Figure 2. Structure of the endometrium

- **The cervix** ensures communication between the uterine cavity and the vagina, it comprises two parts:
 - The ectocervix, the lower part of the cervix in contact with the vagina, is lined with a non-keratinized stratified squamous epithelium.
 - The endocervix, the upper part of the cervix in contact with the uterus, is lined by a simple epithelium which invaginates into the underlying chorion forming the endocervical glands producing cervical mucus.

Cervical mucus moistens the vaginal mucosa, which itself has no glands. It is slightly alkaline (unlike the vaginal environment, which is acidic) and forms a viscous plug that closes the cervix (except at the time of ovulation, when it becomes liquid, "stringy," and loses its function as a plug to allow sperm to pass through).

- **The fallopian tubes** (or fallopian tubes or oviducts) are ducts that extend from the uterus to the ovary. At this point, their fringed funnel-shaped ends, called the pinna, open into the peritoneal cavity opposite the ovary.

- **The ovaries** are attached to the uterus by a ligament. One of the ovaries produces an egg each month, from puberty to menopause.

The ovary can be divided into a cortical zone (ovarian cortex) and a medullary zone (ovarian medulla). In the loose connective tissue of the ovarian medulla are blood and lymphatic vessels, while oocytes are found in the cortical zone at different follicular stages.

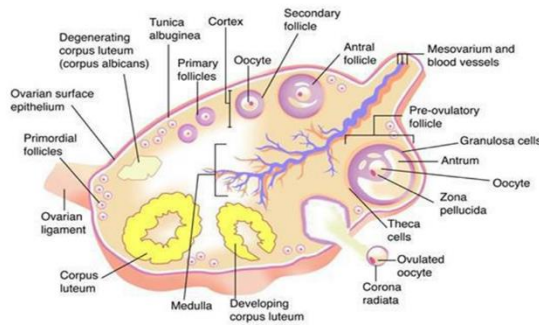


Figure 3. Structure of the ovary

2. PHASES OF OOGENESIS

Oogenesis includes the multiplication, growth, and maturation phases. The growth phase and the beginning of maturation take place within the ovarian follicle and are linked to the evolution of this follicle. The end of maturation is delayed. It is completed after fertilization. There is no differentiation phase.

2.1. Multiplication phase

It concerns oogonia, diploid stem cells and is characterized by a succession of mitoses that will lead to the formation of oocytes I (first order), also diploid. This phase takes place, in women, during embryonic and fetal life. Oogonia are observed in the cortical zone of the embryonic ovary, have a spherical shape and are small in size (15µm), degenerate, for the most part, around the 7th month of intrauterine life (atresia), give oocytes I (2n chromosomes, 2q DNA), larger cells (20 to 40 µm), which immediately after their formation are surrounded by follicular cells and a peripheral membrane that separates them from the rest of the ovarian stroma, the whole designating the primordial follicle, then they begin the first division of meiosis, which stops at the prophase stage. The oocyte then enters a quiescent state in which it can remain for many years (oocyte I blocked in prophase of the first division of meiosis until puberty).

Thus, at the end of this multiplication phase (birth), a non-renewable stock of oocytes I (approximately one million) is constituted, each contained in a primordial follicle.

2.2. Growth phase

It is characterized by a very significant increase in the size of oocyte I, which increases from 20 to 120 µm in diameter. Very long, it only ends at the time of follicle maturation and consists of syntheses of RNA and proteins which will play a crucial role during fertilization and during the early stages of embryonic development.

Primordial follicles regress in large numbers between birth and puberty: only 400,000 will remain at the time of puberty; fewer than 500 will develop until ovulation during the woman's reproductive life.

2.3. Maturation phase

Each month, at the time of ovulation, oocyte I ($2n$ chromosomes, $4q$ DNA) completes the first division of meiosis and gives rise to oocyte II (n chromosomes, $2q$ DNA) with the release of the first polar body. Immediately after, the 2nd division of meiosis begins. But the process is blocked once again (in metaphase of the 2nd division of meiosis) and is conditioned by the occurrence or not of fertilization:

- In the absence of fertilization, the oocyte remains at this stage of meiosis and degenerates after 24 hours.
- If fertilization occurs, oocyte II will complete its maturation and transform into a mature ovum (ovotid) with release of the second polar body.
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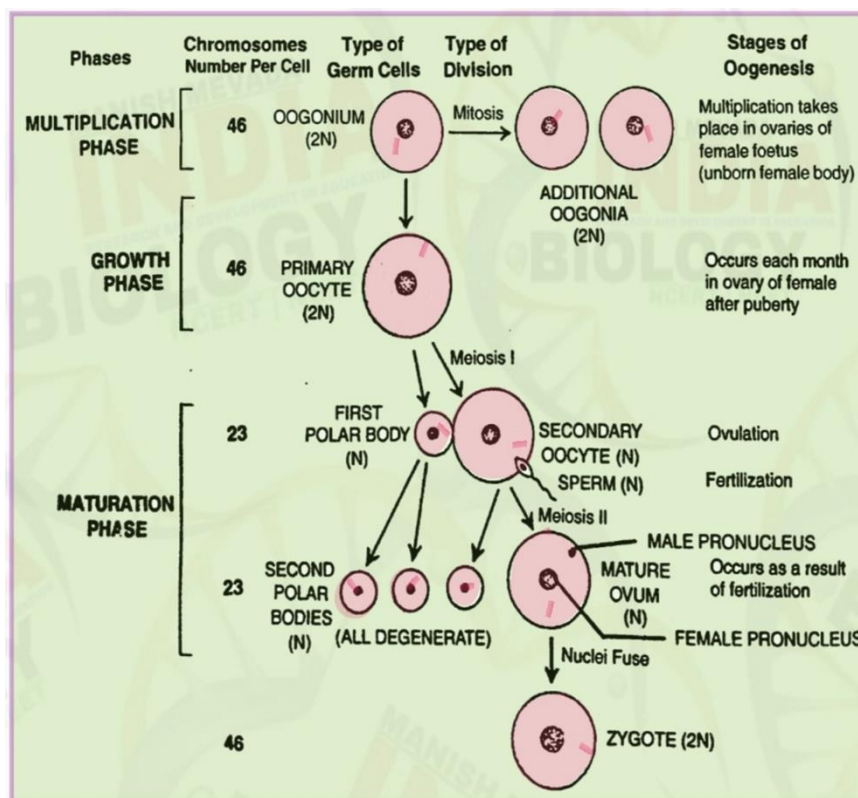


Figure 4. Stages in oogenesis

3. DEVELOPMENT OF OVARIAN FOLLICLES (FOLLICULOGENESIS)

3.1. Primordial follicle

- It is always the most abundant type of follicle on an ovarian section.

- It is a 50 μm diameter sphere that includes an oocyte I and a layer of flattened follicle cells.

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3.2. Primary follicle

- Its diameter increases from 50 to 80 μm . Oocyte I is still stuck in prophase, and the phase of large growth begins.
- The follicle cells become cubic and are arranged in a single layer.

3.3. Secondary follicle (or pre-antral or full follicle)

- Its diameter gradually increases from 80 to 200 μm .
- Oocyte I continues to grow (and reaches 80 μm).
- The pellucid membrane becomes visible under light microscopy (it is a hyaline structure, composed of glycoproteins whose origin is essentially oocyte but the follicle cells produce some of it).
- The follicle cells multiply and arrange themselves in about twenty layers around the oocyte, constituting the granulosa. The innermost layer, regularly arranged around the pellucid membrane, is called the Corona radiata.
- Slavjanski's membrane separates the granulosa from the theca interna, which forms around the basement membrane by differentiation of the cortical stroma.
- The cells, initially spindle-shaped, become cubic.

3.4. Tertiary (or cavitory or antral) follicle

- The follicle cells surround small liquid cavities whose confluence will form the antrum.
- It is defined by the appearance of small cavities within the granulosa which contain a liquid called Liquor folliculi. These cavities are small rosette formations, the "Call and Exner bodies".
- The follicular diameter continues to increase to reach 10 to 15 mm at the end of this stage.
- The oocyte is still blocked in prophase I and reaches 100 μm in diameter.
- The pellucid membrane reaches 15 μm in thickness. Cell multiplication continues. The oocyte is pushed to the side of the follicle. It remains surrounded by a mass of follicle cells constituting the Cumulus oophorus (or Cumulus proliger) which keeps the oocyte attached to the rest of the follicle.
- The external theca forms around the previous one. It is fibrous connective tissue that condenses around the internal theca.

3.5. Mature follicle (Graafian follicle)

- Its diameter is 18 or even 20 mm.
 - The oocyte has a diameter of 120 to 150 μm .
 - The nucleus migrates to the periphery of the cytoplasm and resumes the process of meiosis.
 - Reductional division is completed and is accompanied by the expulsion of the first polar body. The oocyte becomes an oocyte II.
- The zona pellucida increases slightly in thickness.

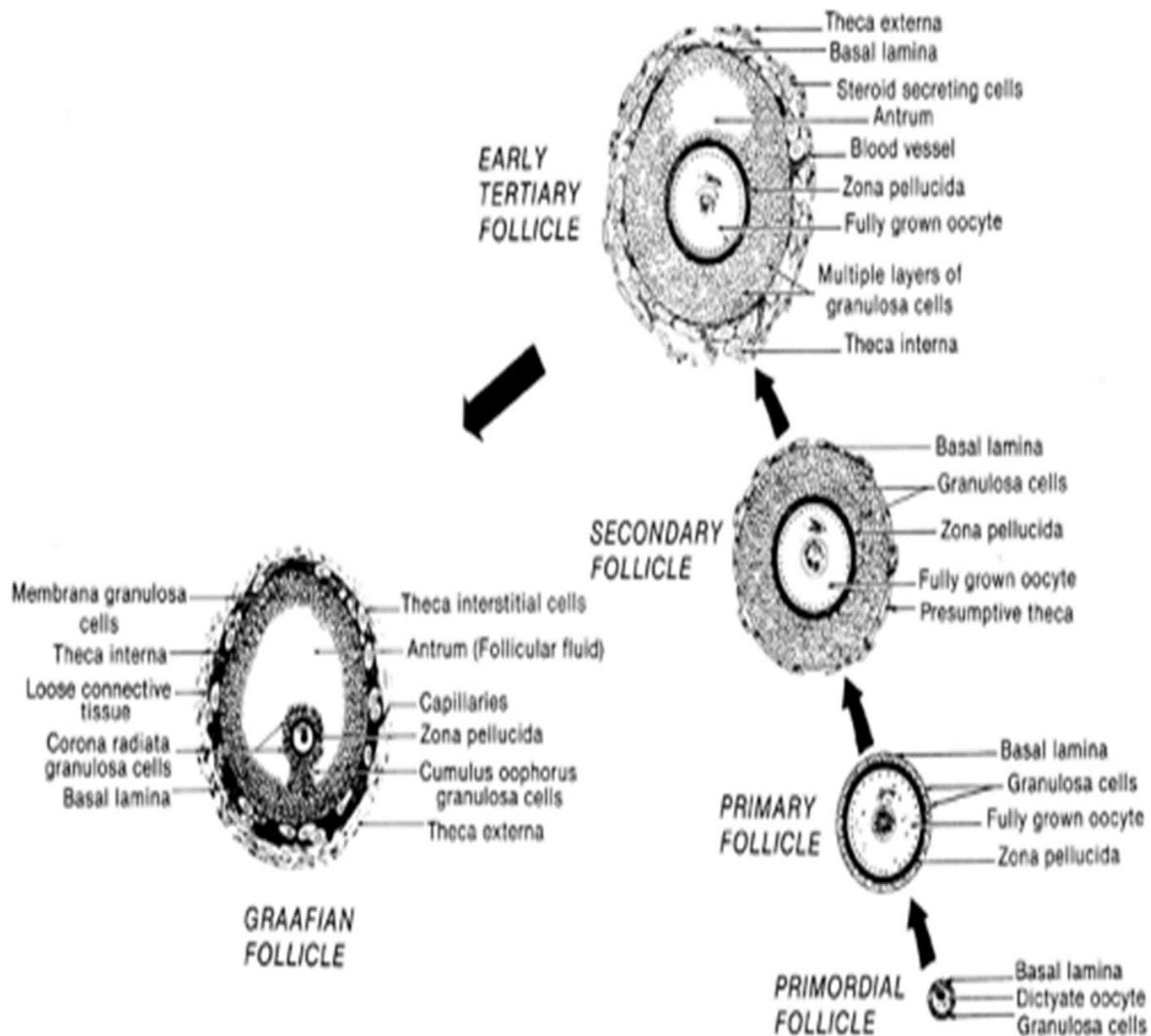


Plate 2. Folliculogenesis

4. BIOLOGICAL CHARACTERISTICS OF THE OVOCYTE

From the oocyte towards the periphery, we distinguish:

The perivitelline space; very small clear space

The zona pellucida: layer 15 to 20 μm thick,

made up of glycoproteins.

Periocyte cells: a layer of contiguous cells forms the Corona radiata and around it are Cumulus cells. The intercellular spaces are wide and occupied by a viscous substance composed of glycoproteins and hyaluronic acid.

The characteristics of the oocyte are summarized as follows:

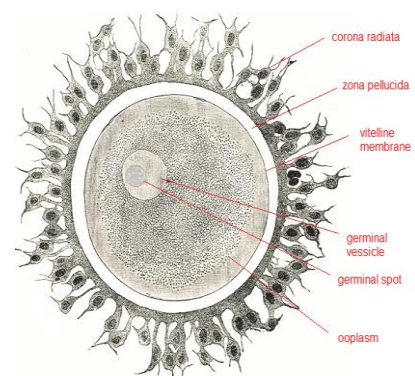


Figure 4. Structure of oocyte II

4.1. Metabolic activity

It is a spherical cell of 150 μm in diameter, relatively inert, without synthetic activity. The nuclear material is blocked in metaphase of 2nd meiotic division. The chromosomes are haploid in number (but with $2n$ DNA). All oocytes are similar (22 autosomes and one X gonosome).

4.2. Vitality

Oocyte vitality is reduced. In the absence of fertilization, the oocyte survives 24 to 48 hours. It ages progressively: the later fertilization occurs, the greater the risk of conceptus abnormalities.

4.1.3. Fertility

The fertilizability of the oocyte (ability to fuse with the male gamete) is conditioned by the degree of cytoplasmic and nuclear maturation (blockage in metaphase II).

5. DIFFERENCES BETWEEN MALE AND FEMALE GAMETOGENESIS

From a genetic point of view, the principle of gametogenesis is the same in both sexes, but there are important differences.

5.1. Kinetics

In women, the proliferation phase occurs exclusively during fetal life, between 3 and 7 months of intrauterine life. The cells remain quiescent, blocked at the end of the prophase of the 1st meiotic division. The number of gametes is fixed before birth. There is incomplete development of a certain number of cells, but gametogenesis only results in the formation of mature gametes from puberty. Gametogenesis stops at menopause, around the age of 50, due to the exhaustion of the stock of germ cells. Thus, some oocytes can remain blocked for nearly 50 years.

In humans, stem cells remain dormant and undifferentiated until puberty. At this point, gametogenesis begins and becomes continuous. The incessant production of sperm generally persists until the individual's death.

5.2. Result of meiosis

In women, during each division of meiosis, one of the cells retains almost all of the original cytoplasm and continues its maturation. The other cell is a polar body. This is a much smaller cell that contains almost only the genetic material from meiosis. The first polar body is expelled during reductional mitosis, while the second, resulting from equational division, is ejected at fertilization.

In humans, a first-order spermatocyte will give rise to 2 identical second-order spermatocytes, then 4 identical spermatids.

5.3. Number of gametes

The female ovary releases approximately 350 oocytes during the active reproductive period. In contrast, male gametogenesis produces 100 to 300 million sperm per ejaculate.

5.4. Characteristics of gametes

The spermatozoon is a small, highly differentiated, isolated, motile cell with little cytoplasm. The oocyte is a very large cell, rich in cytoplasm, immobile and surrounded by specific envelopes.

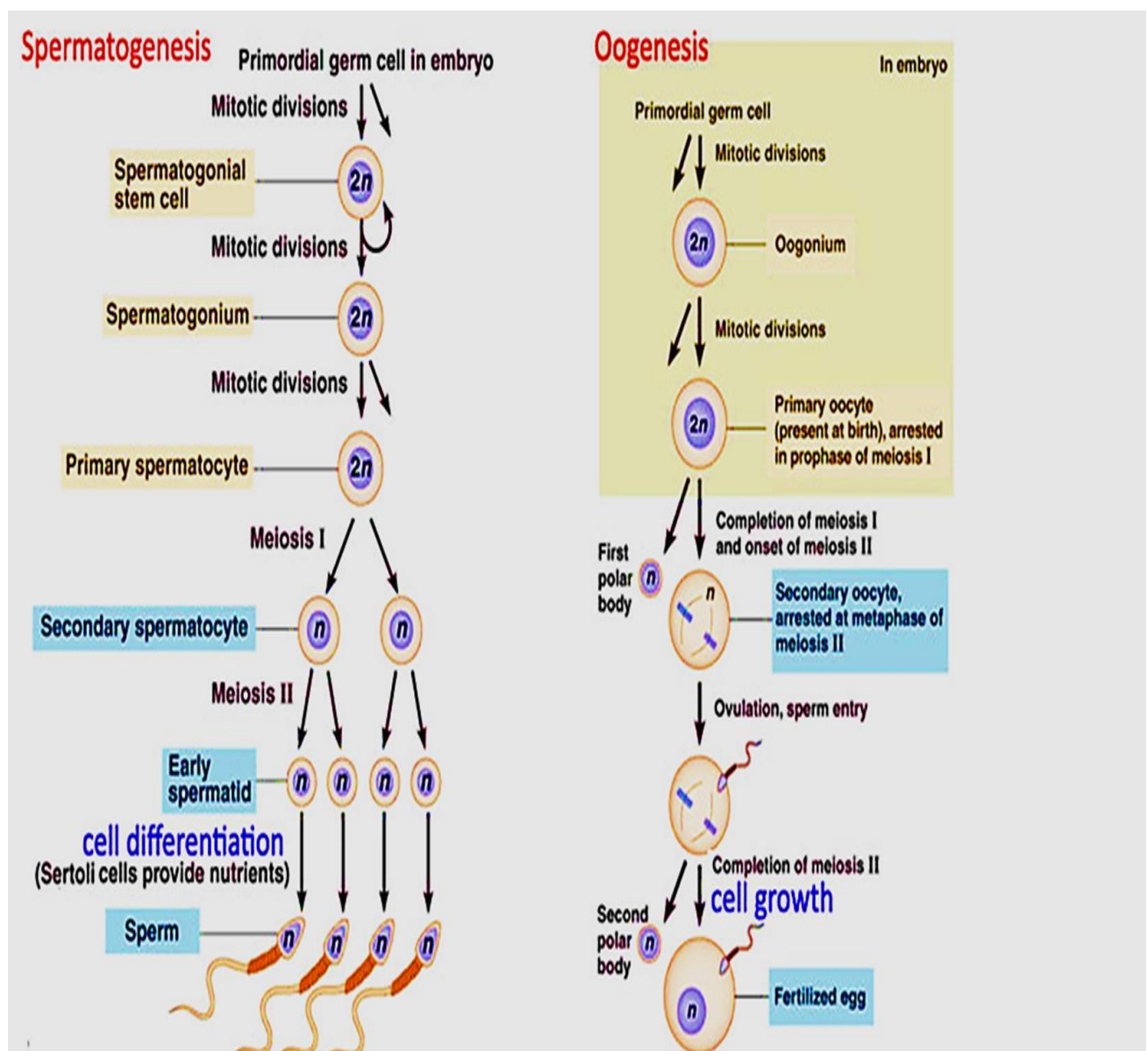


Plate 3. Differences between spermatogenesis and oogenesis